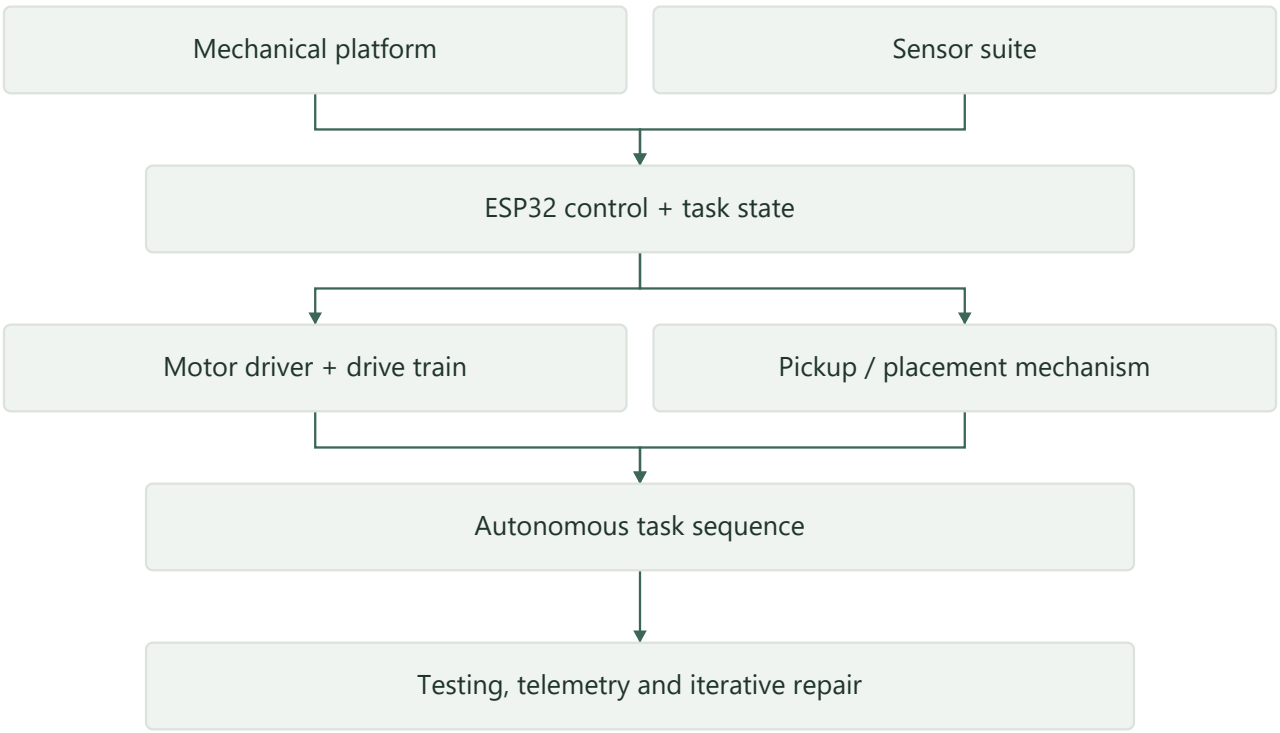
Roblocs

Autonomous robotics · National Robotics League

A physical robotics project combining a fabricated mobile platform, multisensor control, staged testing and autonomous task sequencing, documented through the original team portfolio.

Architecture



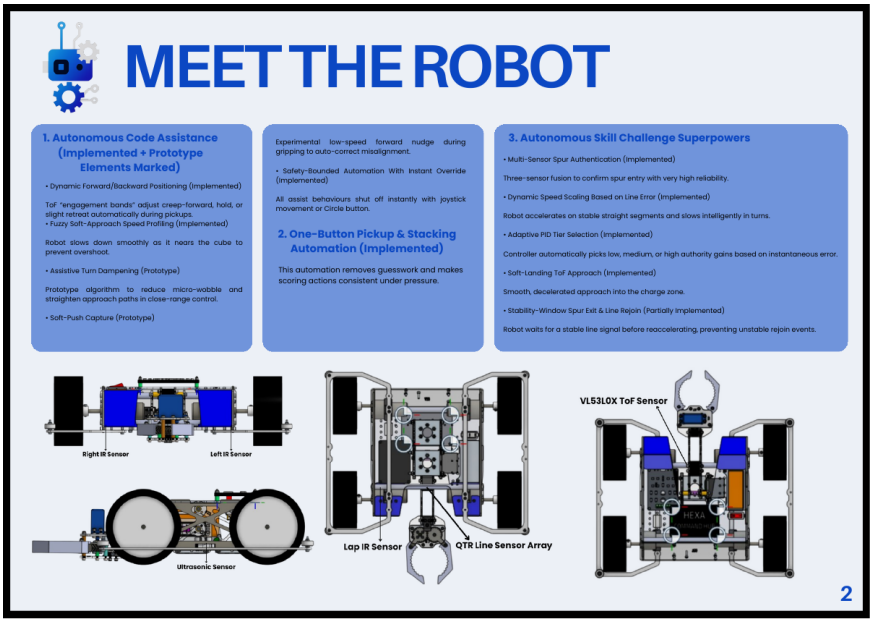
Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Control	ESP32 + motor driver	Sensor integration and actuation
Sensing	Line, proximity, distance and lap sensing	Navigation and task triggers
Mechanism	Original build and mechanism diagrams	Physical integration
Validation	Documented trials and analysis	Team-reported evidence with provenance

Team engineering portfolio

Robot architecture & team responsibility

Roblocs combines a differential-drive BaseBot, articulated gripper, ESP32 control and a distributed sensor arrangement for the National Robotics League. The supplied portfolio now documents a physical build and its staged development, extending the earlier analysis-only snapshot.



Roblocs supplied portfolio, PDF pp. 2-3, 5-6. Team document claims; full firmware/CAD release not supplied.

System integration

The chassis combines polycarbonate plates and printed mounts; a servo operates the arm/gripper through helical gears. The ESP32 coordinates PWM drive, servo commands, PS5 input and sensor processing, with an MDD10 motor driver in the initial build.

Contribution record

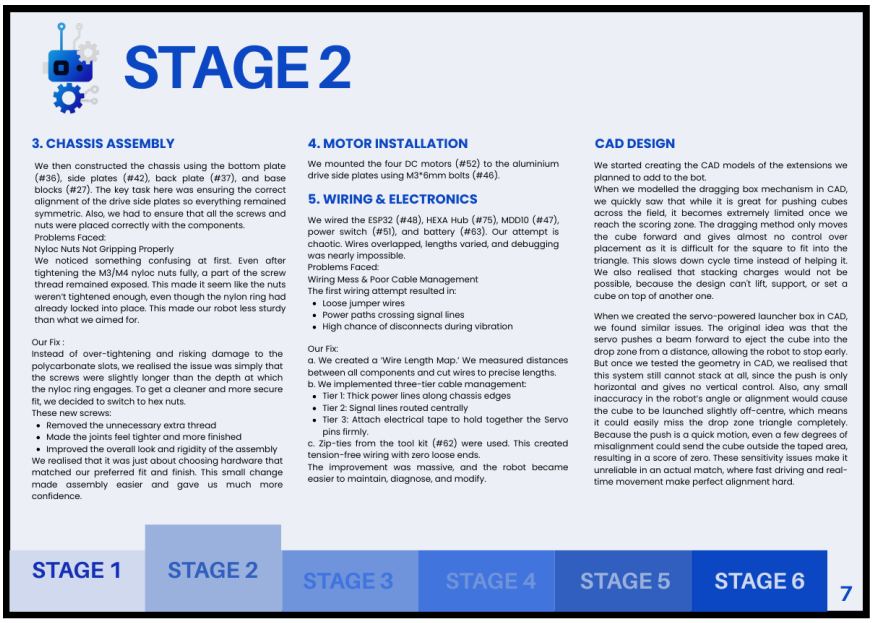
The team charter identifies Arham Sethi as Autonomous Systems Engineer and Project Manager: autonomous-challenge code, main robot codebase, debugging, software/hardware integration, testing and innovation oversight. Other team members own mechanical, CAD, electronics, simulation and brand responsibilities.

Engineering detail	Interpretation
Mechanical path	Drive motors, arm linkage and dual-finger capture
Control path	Manual driving, assist behaviors and autonomous challenge

Source / interpretation: Roblocs supplied portfolio, PDF pp. 2-3, 5-6. Team document claims; full firmware/CAD release not supplied.

Mechanical assembly & failure prevention

The assembly record treats structural alignment and serviceability as engineering problems, rather than assuming the kit is mechanically complete after fastening its parts.



Roblocs PDF pp. 5-9. These are documented build observations; component strength is not independently certified.

Load paths and geometry

The helical gripper transfers servo torque into opposed finger motion; gear alignment, spacers and shaft support affect friction and jamming. A parallelogram arm maintains gripper orientation as the arm moves. The portfolio discusses a 100-150 g charge and 10-15 cm effective lever arm as initial torque-analysis assumptions.

Practical assembly corrections

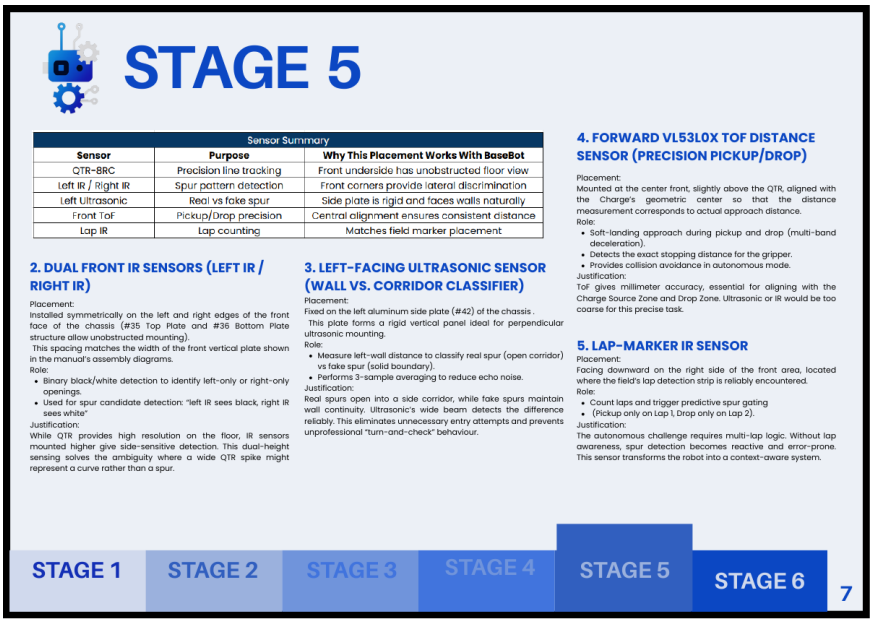
The team describes filing/deburring printed gear teeth, loosening and aligning linkages to restore smooth motion, tightening nuts on M3/M4 fasteners, and organizing wiring by power, signal and analogue lines. A front frame needed clearance correction around motor mounting.

Engineering detail	Interpretation
Risk	Gear binding and chassis twist alter repeatability
Response	Check free motion, parallelism and clearances before powered tests

Source / interpretation: Roblocs PDF pp. 5-9. These are documented build observations; component strength is not independently certified.

Sensor placement & spur authentication

Spur recognition combines floor geometry, asymmetric side signals and corridor distance. The design avoids turning into every suspected spur merely to test it.



Roblocs PDF pp. 17-19. Placement figure is reproduced from the supplied team portfolio.

Complementary observations

The QTR-8RC reflectance array sits on the front underside ahead of the wheels and yields weighted line error plus wide-patch detection. Left/right IR sensors on the front edges discriminate asymmetric openings. A left-facing ultrasonic sensor measures whether the boundary is a wall or an open corridor, using three-sample averaging.

Task-specific sensing

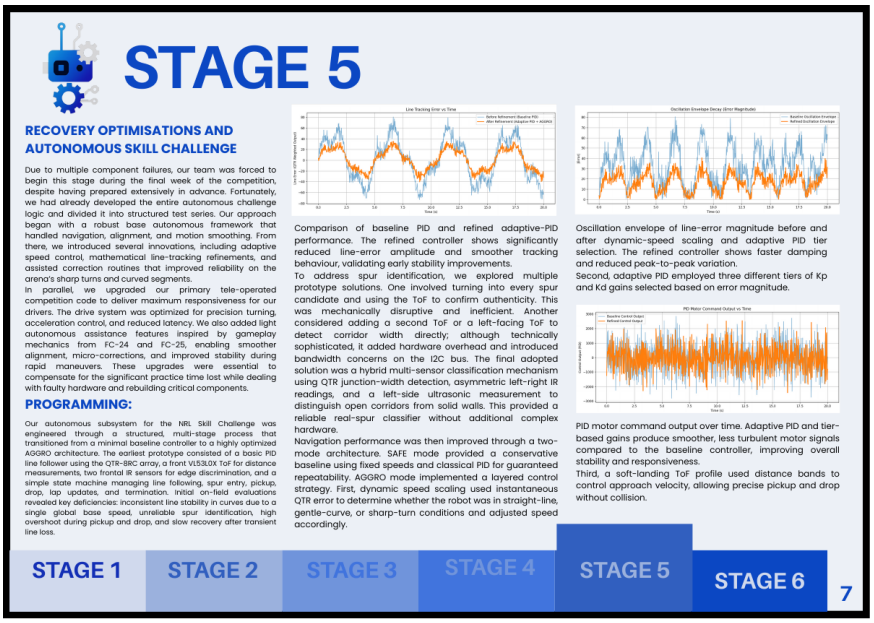
The centered front VL53L0X ToF sensor measures pickup/drop approach distance. A downward-facing lap IR sensor on the right identifies lap markers. This gives the planner both local geometry and task phase; the portfolio describes pickup on lap 1 and drop on lap 2.

Engineering detail	Interpretation
Fusion rationale	No single ambiguous wide QTR patch authorizes a turn
Placement rationale	Floor, side-wall and target-distance channels observe different geometry

Source / interpretation: Roblocs PDF pp. 17-19. Placement figure is reproduced from the supplied team portfolio.

Adaptive PID & autonomous state logic

The autonomous subsystem evolved from fixed-speed PID line following into an error-conditioned navigation architecture, with separate spur entry, manipulation, lap update and recovery behavior.



Roblocs PDF pp. 17-19 and 22. Source status labels differ across pages; retain prototype/partial qualifiers. Charts are portfolio-reported illustrations, not independently audited trials.

Continuous control

Dynamic speed scaling uses instantaneous line error to distinguish straight segments, gentle curves and sharp turns. Low, medium and high PID-authority tiers change gains with error magnitude. The intended effect is fast straight-line travel with reduced oscillation in curves.

Discrete transitions

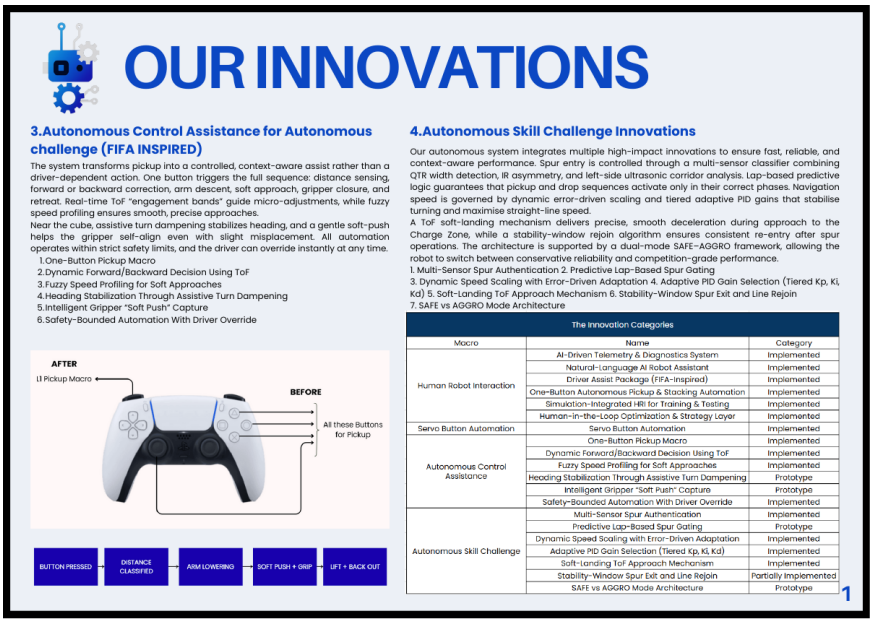
ToF distance bands govern approach deceleration. A stability-window rejoin waits for a consistent line signal before full acceleration after exiting a spur. The portfolio contrasts conservative SAFE behavior with more aggressive AGGRO behavior, but its innovation table labels the dual-mode architecture as a prototype and rejoin as partially implemented.

Engineering detail	Interpretation
Continuous variables	QTR error, derivative, motor output and ToF distance
State constraints	Spur authenticity, lap phase, stable rejoin and timeout

Source / interpretation: Roblocs PDF pp. 17-19 and 22. Source status labels differ across pages; retain prototype/partial qualifiers. Charts are portfolio-reported illustrations, not independently audited trials.

One-button pickup & driver override

The pickup assist reduces a multi-button manual sequence to a bounded macro while preserving driver intervention. It combines target-distance classification with scripted arm and gripper actions.



Roblocs PDF pp. 18, 21-22. The 3.13 s versus 1.73 s stacking bar chart on p.21 is explicitly synthetic predictive data.

Macro sequence

The L1-triggered finite-state macro tracks MacroMode, macroStep and timeouts. ToF classifies TOO_FAR, IDEAL or TOO_CLOSE; the robot moves forward/backward as needed, lowers to ARM_PICK_POS, closes the gripper with a low-speed soft push, lifts to ARM_UP, reverses and sets hasCharge.

Limits and status

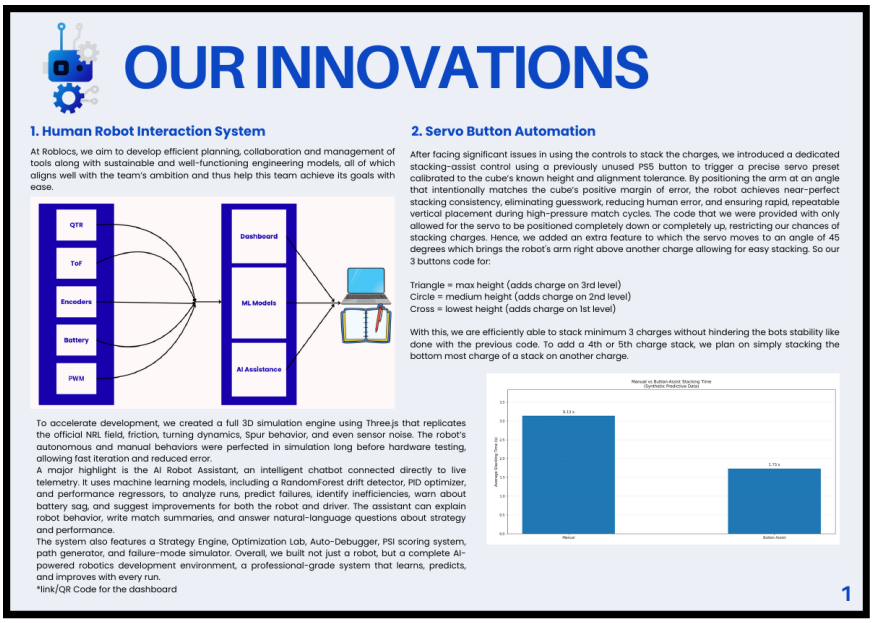
Large stick deflection or Circle cancels the assist; per-step/global timeouts prevent a trapped driver. Triangle/Circle/Cross map to stacking heights in the separate servo automation. Assistive turn dampening and soft-push capture remain marked prototype in the status table, even where narrative wording is more expansive.

Engineering detail	Interpretation
Human control	Manual driving remains the primary intervention path
Automation boundary	Local high-confidence actions rather than vision-based full autonomy

Source / interpretation: Roblocs PDF pp. 18, 21-22. The 3.13 s versus 1.73 s stacking bar chart on p.21 is explicitly synthetic predictive data.

Telemetry, simulation & optimization

The portfolio describes a Three.js simulation and a telemetry-oriented development environment for refining behavior before hardware testing. This is a separate software layer around the robot control loop.



Roblocs PDF pp. 10-11, 13-14, 18, 21. HRI architecture is original supplied diagram; implementation files and run records require later repository linkage.

Data path

The documented telemetry channels include QTR error, motor PWM, ToF distance, battery health, pose and mode. A Python optimization workflow evaluates candidate PID gains against oscillation, overshoot and time-to-stabilize, then returns selected parameters for deployment.

Diagnostic interface

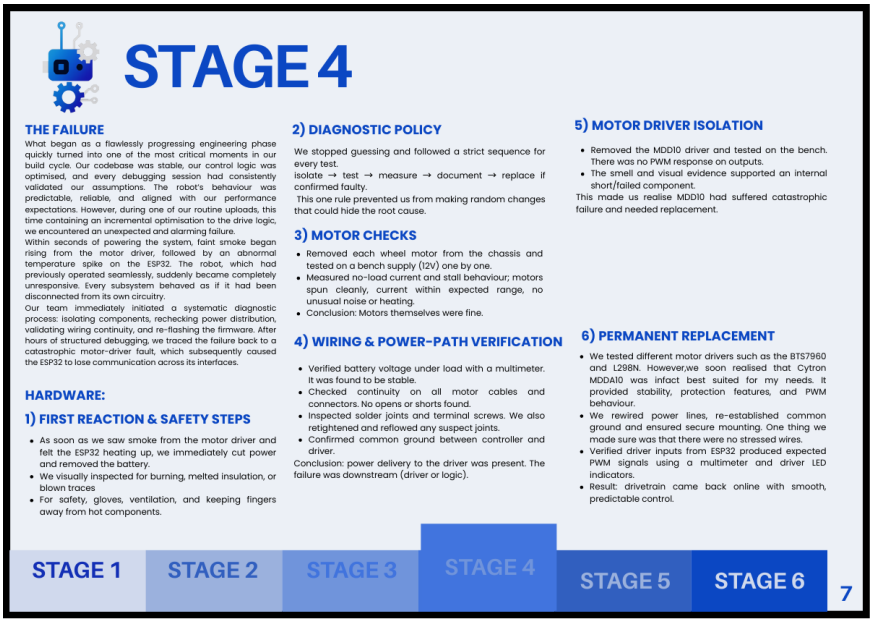
The described assistant combines a RandomForest drift detector, performance regressors and a PID optimizer with natural-language run explanation. Simulation enables fault and scenario tests, but the portfolio explicitly states that early ML models were trained on synthetic samples. Such models should not be represented as validated physical-failure predictors.

Engineering detail	Interpretation
Simulation role	Repeatable scenarios and controlled controller comparisons
Physical role	Check predicted behavior under actual friction, battery and load

Source / interpretation: Roblocs PDF pp. 10-11, 13-14, 18, 21. HRI architecture is original supplied diagram; implementation files and run records require later repository linkage.

Component failure & recovery workflow

The build narrative includes a sudden loss of drivetrain movement, followed by electrical isolation and a replacement motor driver. This is a useful engineering case study in narrowing a fault before changing unrelated code.



Roblocs PDF pp. 15-16. Record describes team diagnosis; no component forensic report is supplied.

Fault isolation

The team stopped actuation, inspected the chassis and motors, verified battery voltage and wiring continuity, then removed the MDD10 and tested it on a bench. The portfolio attributes the final fault to the motor driver and records a replacement with a Cytron MDD30A.

Secondary recovery

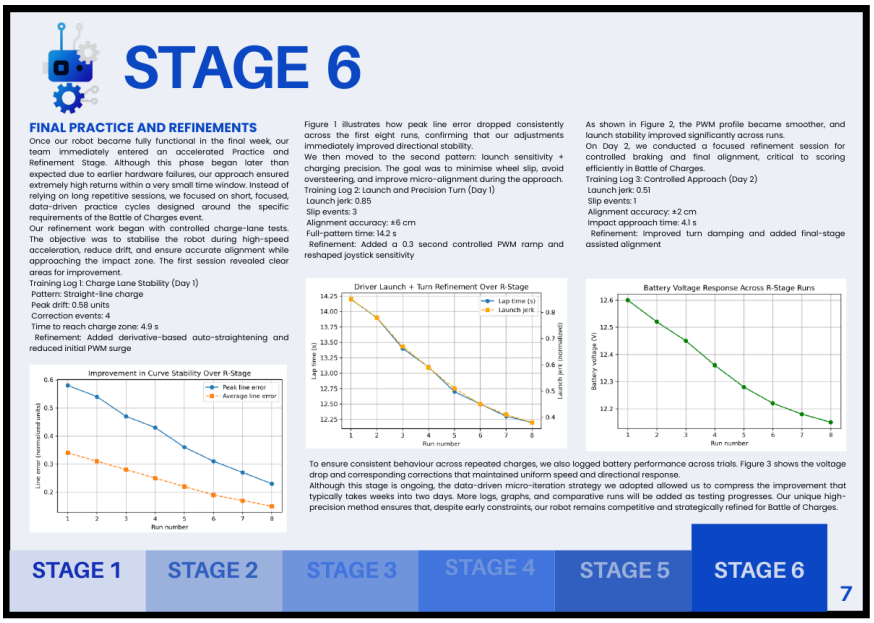
A later ESP32 overheating incident caused loss of robot communication; a replacement board restored operation while a further replacement command hub was awaited. Wiring was rerouted, loose Allen fasteners checked and cable paths organized to reduce interference and accidental contact.

Engineering detail	Interpretation
Diagnostic order	Power and physical checks before assumptions about firmware
Integration lesson	A functioning controller cannot compensate for a failed power stage

Source / interpretation: Roblocs PDF pp. 15-16. Record describes team diagnosis; no component forensic report is supplied.

Test evidence & qualification of outcomes

The final practice stage documents short controlled runs for drift, launch behavior and alignment. These records are valuable as a development narrative, with their measurement definitions and provenance kept visible.



Roblocs PDF pp. 18, 20-22; prior Roblocs plotting scripts. Some statuses and plotting provenance remain inconsistent.

Reported refinement logs

The portfolio lists an initial straight-line run with peak drift 0.58, four correction events and 4.9 s to the charge zone. A later controlled approach lists launch jerk 0.51, one slip event, alignment within ±2 cm and 4.1 s approach time. It also describes a 0.3 s PWM ramp and joystick sensitivity adjustment.

Evidence separation

Several analysis files supplied earlier generate synthetic traces, and portfolio p.18 explicitly calls its ToF curve synthetic. Therefore the portfolio charts should retain their own labels, and numeric improvement claims need original logs and test conditions before being generalized. No national ranking or award is inferred from these pages.

Engineering detail	Interpretation
Record needed	Run ID, firmware hash, battery state, geometry and units
Reason	Distinguish controlled illustration from measured physical comparison

Source / interpretation: Roblocs PDF pp. 18, 20-22; prior Roblocs plotting scripts. Some statuses and plotting provenance remain inconsistent.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Primary record	30-page original robotics portfolio	Supplied PDF
Engineering detail	Mechanisms, control and iterations	Source-page gallery and companion
Attribution	Team work retained	Individual roles only where documented

Selected implementation files

File / record	Path / reference
asc.py	Roblocs/asc.py
g_ler.py	Roblocs/g_ler.py
servobutton.py	Roblocs/servobutton.py
dataset_gen.py	Roblocs/dataset_gen.py

Original 30-page team engineering portfolio and local analysis scripts; the companion now includes the hardware and control evidence supplied in the portfolio.

Evidence boundary

The original portfolio documents the team's system and reported tests. Simulation or illustrative plots are labeled separately from physical measurements; team achievements are not assigned wholly to one contributor.

Technical note

The companion links technical claims and design iterations to original source pages. Controller diagrams, mechanical views and test graphics are reproduced with their existing context.